



Evolution of microstructure, mechanical and corrosion properties of AlCoCrFeNi high-entropy alloy prepared by direct laser fabrication



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ABSTRACT

High entropy alloy (HEA) is an emerging class of engineering materials that shows promising potential for high temperature applications. These multi-component alloys are mostly fabricated by arc melting. In this study, direct laser fabrication (DLF) is utilized to prepare AlCoCrFeNi high entropy alloy at optimized operation parameters. The phase, microstructure, mechanical and corrosion properties of as-deposited alloy as well as samples aged at temperatures of 600 °C, 800 °C, 1000 °C and 1200 °C for 168 h have been investigated. The results show that high cooling rate during deposition inhibits the formation of FCC phase, leading to a nearly single B2 solid solution structure for as-deposited sample. After ageing at 800 °C, 1000 °C and 1200 °C, the microstructures exhibit intergranular needle-like and plate-like FCC phase precipitates and wall shaped FCC phase precipitates along grain boundaries. As the FCC phase is softer than B2 phase, the formation of the FCC phase during ageing results in reduced compressive yield strength accompanied with enhanced ductility. The potential difference between Fe–Cr rich FCC phase and Al–Ni rich B2 matrix means the alloy is susceptible to galvanic corrosion, with the B2 matrix corroding preferentially.

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1. Introduction

HEA is an emerging class of metallic materials, which was proposed by Yeh et al. [1] at Tsing Hua University of Taiwan as far back 1995. Unlike traditional alloys, HEAs consist of several principal elements with equal or near equal-molar fractions. The ability to form simple solid solution phases in HEAs instead of complex multi-phase microstructures consisting of intermetallics provides unique properties and has attracted much research interest in recent years [2–5]. Numerous HEA systems have been synthesized and characterized over the past twenty years [6,7]. Various early parameters were derived from existing HEA systems to predict the formation of solid solution phases [8–10]. Recently, a new parameter, Φ , the ratio between the Gibbs free energy of a completely disordered solid solution to that of the most likely intermetallic or segregated binary system, is proved to be superior for the prediction of single phase HEAs [11].

Compared with traditional alloys, HEAs exhibit high strength, excellent resistance to thermal softening, and good structural

stability at elevated temperatures [12–16], which makes them promising materials for high temperature applications. One of the most comprehensively investigated high entropy alloy systems is Al_xCoCrFeNi, due to the relative abundance and low-cost constituent elements coupled with excellent mechanical properties. The microstructure of Al_xCoCrFeNi can be tailored from single FCC solid solution, duplex FCC plus BCC solid solutions to single BCC solid solution with increasing aluminium mole ratio, meanwhile the hardness is increased from 116 Hv to 509 Hv [17,18]. There are not many publications about tensile properties of this HEA system [19] and most compression results available are from AlCoCrFeNi, exhibiting impressive 1.37–1.45 GPa yield stress, 2.96–3.53 GPa ultimate stress and 15.5–24.5% strain [20,21]. Although phases and microstructures at elevated temperatures have been characterized for some compositions in the Al_xCoCrFeNi system [22], the influence of phase and microstructure evolution on the mechanical and electrochemical properties has not been examined in the literature. In the present study, the AlCoCrFeNi alloy was selected to examine the microstructure and phase evolution at high temperatures and their effects on the compression and corrosion properties.

Various processing routes such as arc melting [4,17,22,23], spark plasma sintering [24–27], mechanical alloying [27–29], plasma

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spray deposition [30], sputter deposition [31] and laser cladding [32,33] have been utilized to fabricate HEAs. DLF is a rapid prototyping technology based on laser cladding that can produce fully-dense metallic parts with complex near-net shape by laser melting and deposition of powders layer by layer. Substantial research work has been done on direct laser fabricated alloys using pre-alloyed powders [34–37] or pure elemental powder blends [38–40]. With the use of elemental powder blends and controlled supply from several hoppers, DLF provides an alternative process to prepare in situ homogenous HEAs efficiently and effectively. In recent years, there have been only a small number of publications regarding DLF of HEAs [41–43]. In the present study, DLF has been employed to prepare AlCoCrFeNi samples.

2. Materials and methods

HEA samples with nominal composition AlCoCrFeNi were prepared by DLF. A TRUMPF TruLaser Cell 7040 apparatus and spherical gas-atomised elemental powder blends were used. The powders of Al, Co, Cr, Fe, and Ni were supplied by American Elements and Micron Metals with 99.99% purity and 50–150 μm particle size distribution. The equi-atomic CoCrFeNi powder mixture was prepared by carefully weighing the powders using METTLER TOLEDO weighing balance with an accuracy of ± 0.0001 mg. The mixed powder was homogenised in a rotary tumbler for 12 h. Powder feeder system consisting of two hoppers was employed to separate high density CoCrFeNi powder mixture and low density aluminium powder. By adjusting the spin speed of disks in two hoppers, required amount of powders were delivered into the melt region to get the designed composition. The depositions were implemented on a 15 mm Inconel 618 substrate. To minimise oxidation, the deposition process was undertaken in a sealed, controlled environment which was purged with continuous flow of argon gas to maintain the oxygen concentration as low as 10 ppm. Pre-scan was performed three times in order to minimise the thermal stress between deposit and substrate. All deposit columns of $14 \times 14 \times 70$ mm were fabricated at an operating power of 800 W, a beam diameter of 3 mm, a scan speed of 800 mm/min and a layer thickness of 0.25 mm.

Heat treatments were conducted at 600 °C, 800 °C, 1000 °C and 1200 °C for 168 h respectively. The samples were vacuum encapsulated in quartz tubes to avoid oxidation and water quenched to conserve the microstructures. X-ray diffraction profiling was performed to determine the phase constituents of HEA systems. The crystallographic phases were identified by Philips X-ray diffractometer using monochromatic Cu K α ($\lambda = 1.54060$ Å) radiation at 40 kV and 25 mA with a scanning step size of 0.01° and a scanning range of $20^\circ < 2\theta < 120^\circ$. EBSD phase map was obtained by JEOL JSM-7001F FEGSEM. The microstructures of HEAs were characterized by FEI Nova Nano SEM. The back scattered images (BSE) of samples were captured at an operating voltage of 20 kV, working distance of 5 mm, high current mode and 60 μm aperture. Bright field images and selected area diffraction patterns (SADP) were obtained using FEI Tecnai G2 T20 TWIN TEM. The samples for the observation were prepared by cutting a 3 mm diameter disk and grinding it with a dimple grinder (Gatan 656) to form a thin film. Next, thin foils were prepared by ion-beam milling (Gatan 691 PIPS). The local chemical composition of samples was analysed by 3 dimension atom probe (3DAP) and Bruker Quantax 400 EDX equipped with SEM. Needle samples for 3DAP were prepared by electrical discharge machining (EDM) into a square rod of dimensions $0.5 \times 0.5 \times 10$ mm followed by electropolishing the square rod to a sharp tip in a solution of 33% nitric acid at a voltage of ~ 14 V, and a second-stage electropolishing in a solution of 5% perchloric acid at 17–27 V. 3DAP analysis was conducted in

ultrahigh vacuum ($< 10^{-8}$ Pa) and ultralow temperature (60 K) on an Oxford nanoScience energy-compensated 3DAP, at a evaporation rate of 0.01 ion per pulse with the pulse repetition rate of 20 kHz and pulse fraction of 20%. Software PoSAP and IVAS were used to analyse the data.

Cylindrical compression test samples were prepared by EDM. The dimension of the samples was 4 mm in diameter and 6 mm in height (aspect ratio = 1.5). The compression test was performed using Instron 150 kN model 5984 at a constant strain rate $1 \times 10^{-4} \text{ s}^{-1}$ at room temperature. Three tests were conducted for each heat treatment condition. Fracture surfaces of samples tested were analysed using FEI Nova Nano SEM.

A three electrode flat-cell (Princeton Applied Research) was used for the potentiodynamic polarization tests to investigate the effect of phases and microstructures on the corrosion kinetics and mechanism. Test solution was 0.6 M NaCl and test temperature was ambient temperature ($\sim 25^\circ\text{C}$). The reference electrode was a saturated calomel electrode (SCE) and the auxiliary electrode was a Pt-mesh electrode. Polarization experiments were performed via a BioLogic VMP3 potentiostat using EC-Lab 10.02 software. The quasi-steady-state time for an open circuit was 600 s. Scan speed was 1 mV/s from an initial potential of -0.5 V to a final potential of 1.5 V versus the OCP. Each sample was tested at least 6 times. After polarization test, the morphologies of the corroded surfaces were analysed by FEI Nova Nano SEM.

3. Results and discussion

3.1. Phase and microstructure evolution

Fig. 1 exhibits the X-ray diffraction profiles of as-deposited and aged samples. Two cubic crystal structures, face-centred cubic (FCC) and body-centred cubic (BCC) are identified. The BCC solid solution is the dominant phase for as-deposited and 600 °C aged sample. A minor (111) reflection of FCC structure appears for both samples. The FCC solid solution phase is more unambiguously identified after ageing at 800 °C and above, as the intensities of (111) reflection peaks of FCC structure are increased to be comparable to the (110) reflection peaks of BCC. Meanwhile, (200) and (220) reflections of FCC structure appear. A minor (100) reflection of ordered BCC (B2 superstructure) near $2\theta \approx 31^\circ$ appears in all of samples. This indicates two possibilities of BCC structure: disordered BCC phase plus B2 phase or single B2 phase. Clarification will be made in the following investigation with TEM. Besides, σ phase appears after aging at 800 °C, but it is dissolved at higher temperatures. It is evident that there is peak shift of both BCC and FCC reflections after ageing. The shift of FCC reflection is towards low-angle side with increasing ageing temperature while the shift of BCC reflection is towards high-angle side, which is also illustrated by lattice parameters of FCC phase and BCC phase in Table 1 calculated according to Bragg's Law.

The microstructures of as-deposited and aged samples are exhibited in Fig. 2. The images of the specimens were obtained at the cross-section perpendicular to the building direction. As-deposited sample and 600 °C aged samples (Fig. 2a and b) exhibit similar near single BCC solid solution phase structure. Fig. 3 shows the EBSD phase map of 600 °C heat treated sample. In the phase map, red and blue areas indicate the B2/BCC and FCC phases respectively. There is only approximate 2% volume fraction of FCC phase sitting at grain boundaries. Periodic, fine scale structures consisting of alternating bright and dark interconnected phases appear inside grains for both conditions (Fig. 2f and g). These two interconnected phases in 600 °C heat treated sample were investigated by TEM analysis (Fig. 4). SADPs picked from matrix and precipitate phases have exactly the same B2 diffraction pattern,

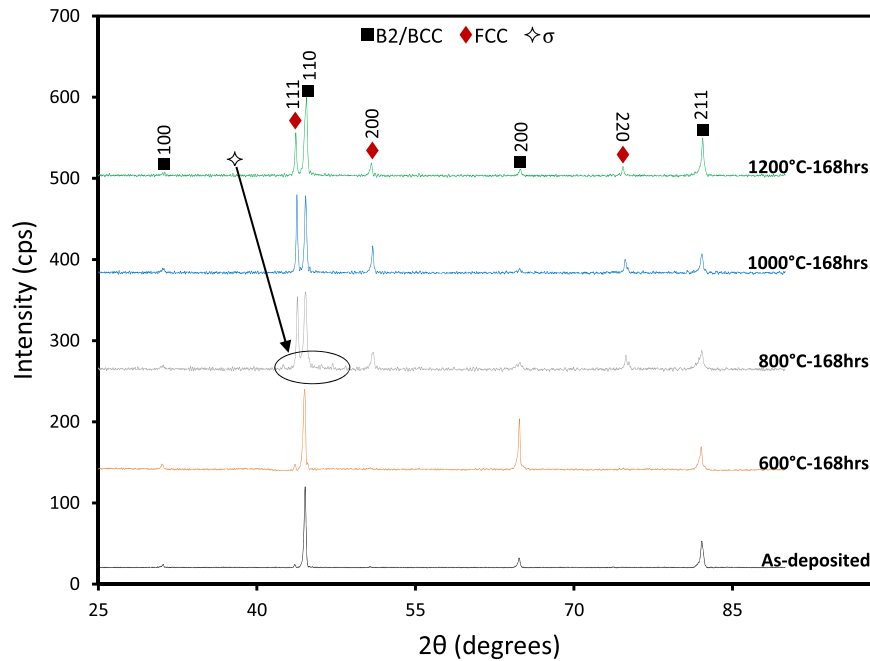


Fig. 1. XRD spectrum of as-deposited and aged AlCoCrFeNi samples.

Table 1

Lattice parameters of as-deposited and aged AlCoCrFeNi alloys.

Sample	Lattice Parameter (Å)	
	FCC	BCC
As-Deposited	—	2.874
Aged 600 °C-168 h	—	2.875
Aged 800 °C-168 h	3.579	2.872
Aged 1000 °C-168 h	3.583	2.872
Aged 1200 °C-168 h	3.591	2.869

which confirms that both of them are B2 solid solution phases. These two phases are assumed to be conjugate B2 phases formed by spinodal decomposition, which differ in composition but have the same crystal structure. The composition difference is demonstrated by three dimensional atom probe tomography of as-deposited specimen (Fig. 5). The segregation of Ni, Al and Cr is extreme. The dark phase is rich in Al and Ni with homogeneous distribution, while the bright phase is rich in Cr. The distribution of Cr is not homogeneous neither in bright phase nor in dark phase, forming clusters of spiral shape and plate shape respectively. Although Fe and Co is distributed more homogeneously compared with the other three elements, it is still evident the bright phase has a higher concentration of these elements than the dark phase.

Samples heat treated at 800 °C and 1000 °C have fairly similar microstructures. The dark phase is the B2 solid solution phase and the bright phase is the FCC solid solution phase. A significant amount of FCC phase is precipitated along grain boundaries with plate morphology and inside grains with thinner plate shaped and needle shaped morphology. The FCC phase precipitated inside grains of the sample aged at 800 °C shows much finer structure than that of 1000 °C aged sample. The volume fractions of FCC phases in these two samples are comparable to those of B2 phases, about 48.4% and 45.8% for 800 °C and 1000 °C heat treated samples respectively. According to the EDX results in Fig. 6, Fig. 7 and Fig. 8, it is found that FCC phase is enriched in Cr, Fe and Co while B2 phase is enriched in Al and Ni. The non-uniform distribution of Fe and Co after ageing is much more evident than that in as-deposited

samples. Less FCC phase (~15%) is formed in the sample heat treated at 1200 °C. Meanwhile, most of the FCC phase is formed along grain boundaries with coarsened plated morphology. Only a small amount of FCC phase is precipitated inside grains. The reason is that with higher aging temperature, the rate of diffusion of Fe, Co and Cr increases, resulting in the growth and coarsening of the FCC precipitates.

The crystal structure and microstructure for as-deposited sample are almost identical to as-cast AlCoCrFeNi prepared by arc melting in literature [17,22,26]. Spinodal decomposition that appears in supersaturated solid solution has been widely observed in high entropy alloys $\text{Al}_x\text{CoCrFeNi}$ and similar compositions prepared by arc melting. It is assumed to be the mechanism that leads to the formation of two conjugate phases [17,18,22,43], as is the case in this work. For some compositions like $\text{AlCoCrFeNiTi}_{0.5}$, the XRD spectrum shows the reflection of two conjugate BCC phases split into two due to greater difference of lattice parameter as a result of composition variation [44]. Zhang et al. [45] calculated isopleth of the $\text{Al}_x\text{CoCrFeNi}$ alloys with $x = 0-3$ above 1000 °C using the CALculation of PHase Diagrams (CALPHAD) approach. The crystal structures and microstructures obtained in this work are not consistent with the calculated isopleth. In Zhang's calculation, disordered BCC solid solution phase exists from 1000 °C to above 1200 °C and the FCC phase is expected to be fully dissolved at 1200 °C. The microstructure and crystal structure observed in this work coupled with DSC heating curves in Wang Woei-Ren's work [22], which demonstrates that FCC phase persists until melting, indicate that the predicted isopleth may not be correct, possibly because the prediction was based on AlCoCrFeNi data extrapolated from the lower order constituent binary and ternary systems.

According to Hume-Rothery rule for binary solid solution systems, small atomic size difference and the identity of crystal structures for two elements facilitate high degree solubility, which can be used to explain element segregation in AlCoCrFeNi. Al and Ni atoms are FCC structure while Fe and Cr atoms are BCC structure. In addition, Al-Ni atomic pair has the smallest atomic size difference compared to Al-Fe/Cr/Co atomic pairs. In terms of thermodynamic aspect, the Al-Ni atomic pair has the lowest mixing enthalpy

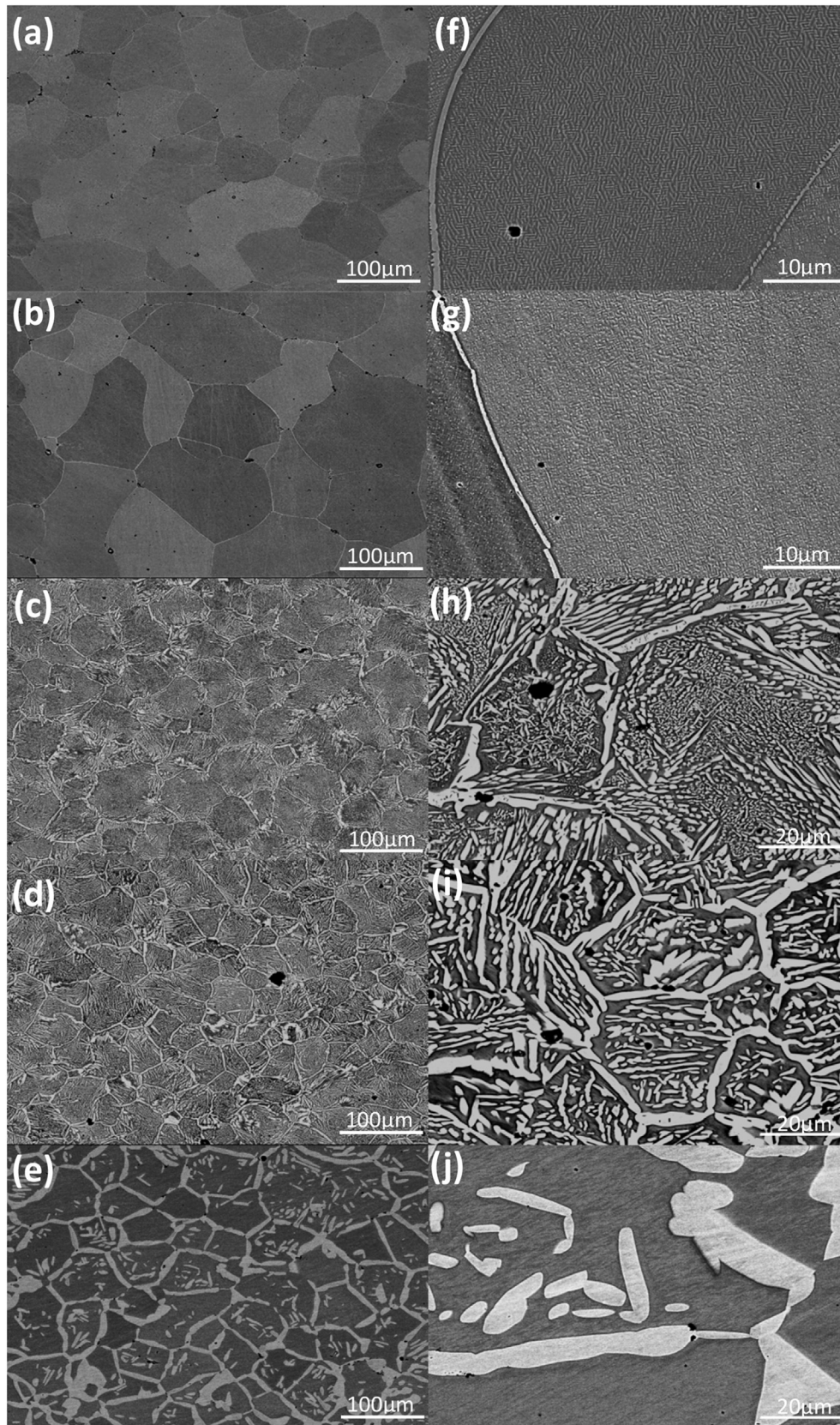


Fig. 2. Back-scattered images of AlCoCrFeNi specimens obtained at the cross-section perpendicular to the build direction with various heat treatment conditions: (a), (b), (c), (d), (e) for as-deposited, aged at 600 °C-168 h, 800 °C-168 h, 1000 °C-168 h and 1200 °C-168 h respectively. (f), (g), (h), (i), (j) are higher magnification corresponding to (a), (b), (c), (d), (e).

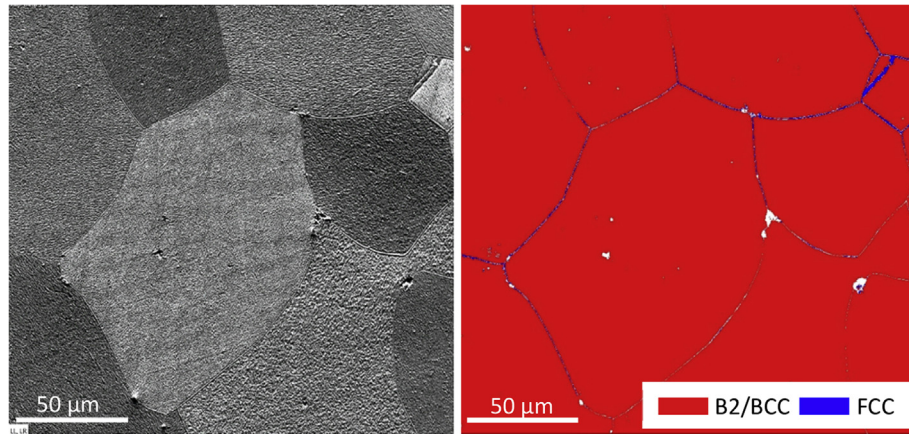


Fig. 3. EBSD phase map of 600 °C aged specimen obtained at the cross-section perpendicular to the build direction.



Fig. 4. Bright Field TEM image of 600 °C aged specimen obtained at the cross-section perpendicular to the build direction and diffraction pattern taken from matrix and precipitate. The electron beam incidence direction of each phase is parallel to $\langle 100 \rangle$.

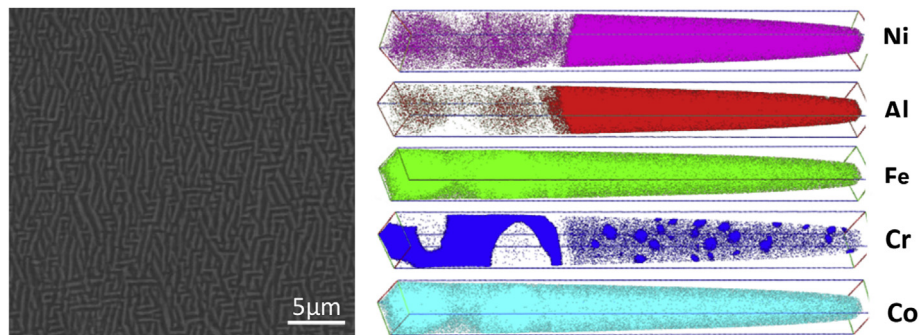


Fig. 5. Three dimensional reconstruction of Al, Co, Cr, Fe and Ni atom positions in AlCoCrFeNi aged at 600 °C for 168 h.

compared to Al-Fe/Cr/Co and Ni-Fe/Cr atomic pairs [46]. Al is known to be a BCC stabilizer in transition metal (TM) based HEAs. The enthalpy of FCC and BCC phases at 800 °C in Al-TM equiatomic binary systems were calculated using Thermo-Calc software. The results indicate Al-Ni atomic pairs have lower enthalpy for BCC structure than FCC structure, which results in the formation of BCC phase at elevated temperatures. The simulations undertaken by Middleburgh et al. [47] predicts that in the CoCrFeNi alloy, Cr is not thermodynamically stable and prefers to segregate into Cr metal cluster. The result of 3D atom probe tomography of Cr behaviour in

this study can be explained by these simulations and shows the prediction to be correct.

3.2. Mechanical properties

Fig. 9 shows the compressive true stress-strain curves of aged AlCoCrFeNi samples. The detailed compression properties are listed in Table 2. The samples aged at 600 °C exhibit excellent room temperature compressive properties. The yield stress observed was 1.31 GPa and the fracture stress was 2.60 GPa, which are superior to

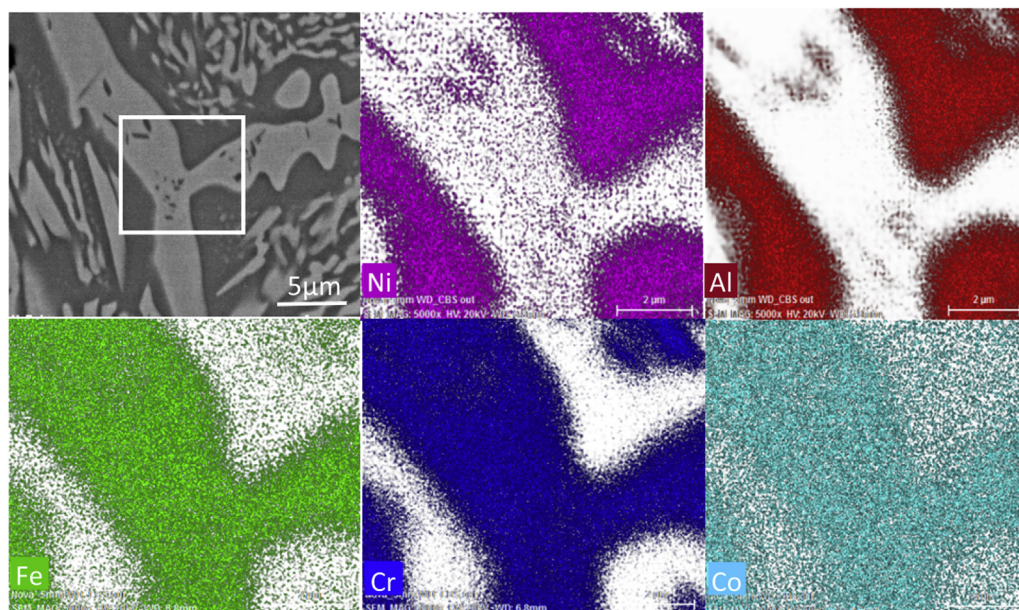


Fig. 6. EDS map of AlCoCrFeNi aged at 800 °C for 168 h.

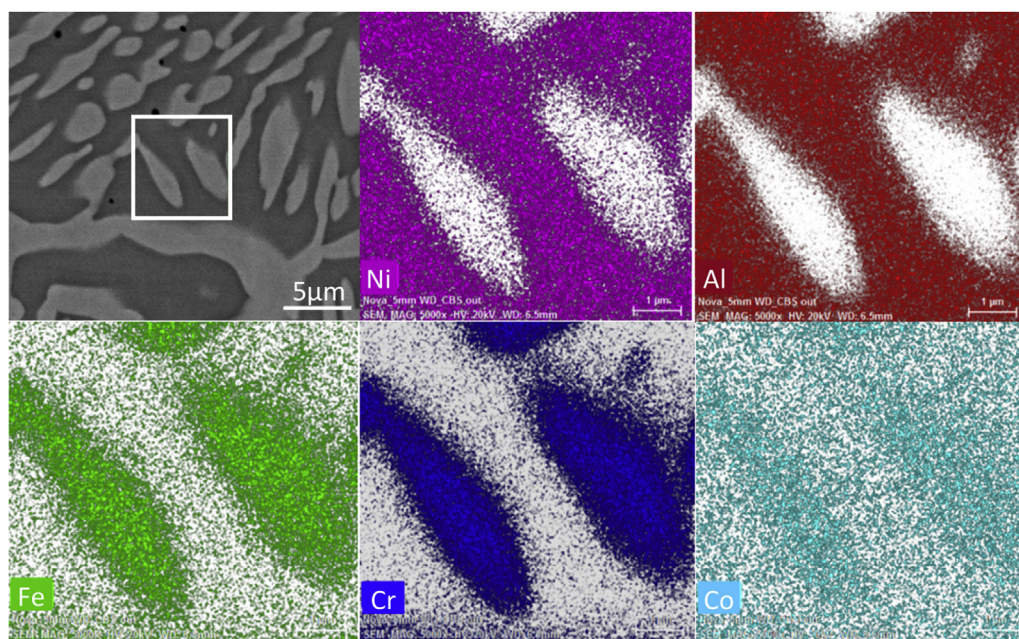


Fig. 7. EDS map of AlCoCrFeNi aged at 1000 °C for 168 h.

some of the high temperature alloys such as nickel based super alloys. Strain hardening is observed for all aged samples. Compared with the sample aged at 600 °C, samples aged at 800 °C, 1000 °C and 1200 °C exhibit improved ductility with plastic strains all above 20%, which results in enhanced propensity for strain hardening and higher fracture stress. However, the yield stresses of samples aged at these three temperatures are reduced compared to the sample aged at 600 °C.

Analysis of fractographs was conducted using SEM to develop a better understanding of the mechanical behaviour. Fig. 10 reveals the morphologies of the fractographs of the fractured samples for different ageing treatments. From the lower magnification images

(Fig. 10a, c, e, g), it is evident that normal fracture occurs, as the fracture direction is along the compressive load. In addition, multiple cracks were observed on the fracture surface. The rough facets, cleavage steps and river like patterns can be observed in Fig. 10 (b) (c), which indicate a typical cleavage fracture for 600 °C and 1200 °C aged samples. Regions consisting of high-density dimple like structures surrounded by cleavage steps and patterns are observed for 800 °C and 1000 °C aged samples (Fig. 8d, f), which indicate a mixed fracture mechanism.

The high strength of this alloy is mainly attributed to solid solution strengthening. As mentioned before, the difference in atomic size of elements results in severe lattice strain and stress. On the

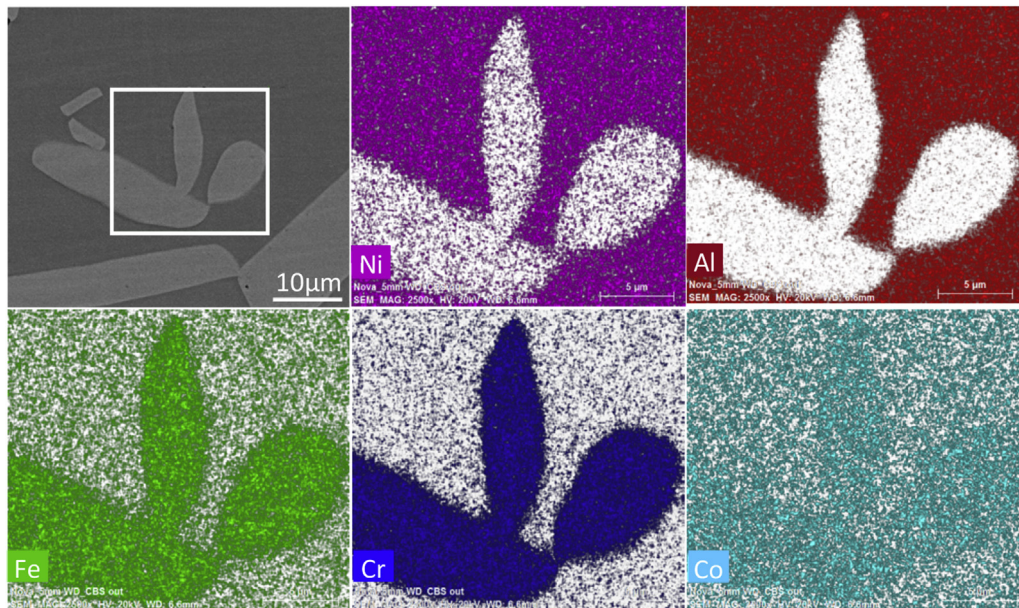


Fig. 8. EDS map of AlCoCrFeNi aged at 1200 °C for 168 h.

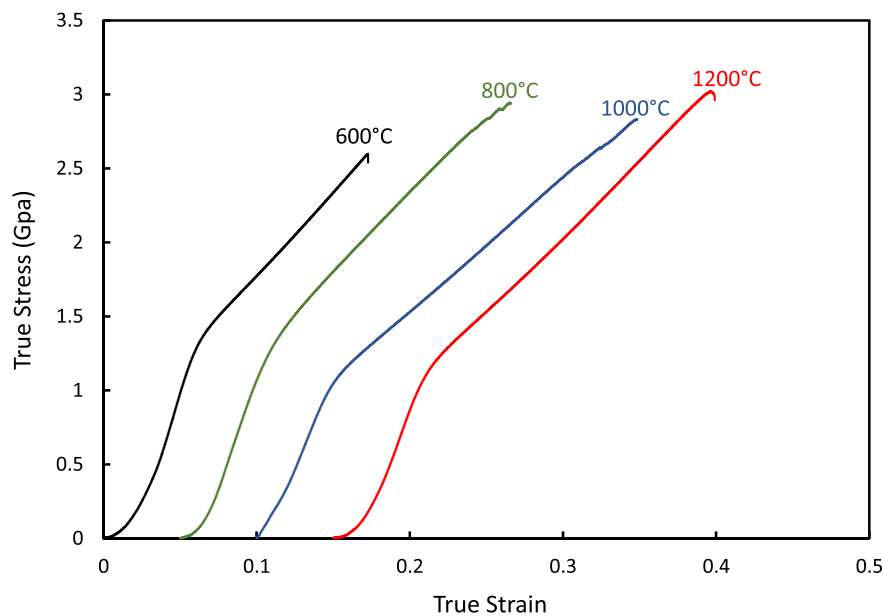


Fig. 9. Compression true stress-strain curve of AlCoCrFeNi aged at various temperatures for 168 h.

Table 2
Room temperature compression properties of AlCoCrFeNi alloys.

Ageing cycle	σ_y (GPa)	σ_{max} (GPa)	ϵ_p (%)
600°C/168 h	1.31	2.60	16.8
800°C/168 h	1.17	2.94	21.6
1000°C/168 h	1.07	2.83	24.9
1200°C/168 h	1.13	3.02	24.2

other hand, the binding force and crystal structure tendency among component elements vary depending on surrounding elements. These non-uniform and asymmetric bindings may lead to extreme lattice distortion and inhomogeneous stress fields throughout the sample. These stress fields interact with dislocations, impeding

their motion and causing an increase in the yield stress of HEAs. Increase in strength caused by solid solution strengthening is also related to the concentration of the solute atoms. For HEAs, each atom can be considered as a solute atom. Therefore, it is anticipated that this alloy would show substantial solid solution strengthening.

In general, it is believed that FCC and FCC-derivative structure based HEAs are softer than their BCC counterparts, as the close-packed nature of FCC structure and more slip plains available for deformation to occur [7]. The crystal structure with more slip plains results in lower lattice friction during dislocation motion and higher ductility. There are 48 slip systems in FCC structure, which is more than that of the BCC structure (12 slip systems). Besides, solid solution hardening could be weaker in FCC phase than the B2 phase in this case. The atomic size difference between Fe, Cr and Co

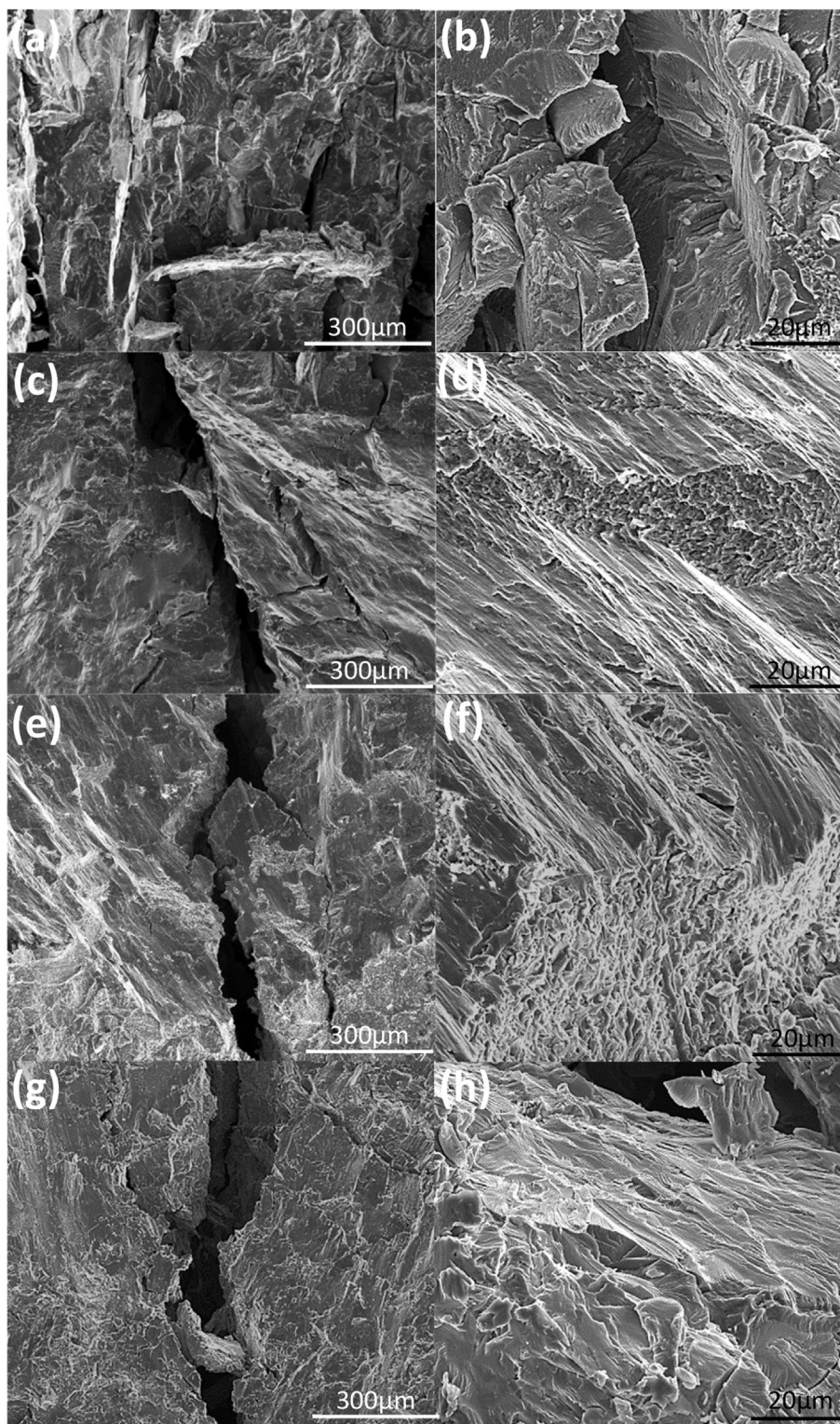


Fig. 10. SEM images of the fracture surfaces of AlCoCrFeNi aged at 600 °C (a and b), 800 °C (c and d), 1000 °C (e and f) and 1200 °C (g and h) after compression test.

enriched in the FCC phase is marginal, while the atomic size difference between Al and Ni enriched in the B2 phase is substantial. Furthermore, the stress fields caused by non-uniform and asymmetric bindings in FCC phase are more homogenous as the binding forces among Fe-Fe, Cr-Cr, Co-Co and Cr-Fe-Co pairs are comparable. Hence solid solution hardening is weaker in FCC phase than the

B2 phase. These would explain the observation that the yield stress of the alloy is reduced while the ductility is improved after precipitation of FCC in 800 °C, 1000 °C and 1200 °C aged samples. The band regions with dimple-like structure on the fracture surfaces of 800 °C and 1000 °C aged samples were expected to be attributed to the more ductile FCC phase. However, they are not dimples simply

formed by ductile fracture mechanism with void coalescence and slip. The dimples appear more like rosettes and tear ridges formed by quasi-cleavage [48], which appear to develop by the formation of micro-cracks and then tearing (plastic deformation) of the connecting material. These dimple regions appear to be formed by quasi-cleavage or a combination of slip and quasi-cleavage.

3.3. Corrosion

The electrochemical corrosion properties of AlCoCrFeNi samples aged at 800 °C, 1000 °C and 1200 °C were evaluated using a potentiodynamic polarization method in 0.6 M NaCl solution. The polarization curves are shown in Fig. 11 and the measured data of electrochemical parameters are listed in Table 3 compared with AISI 304L stainless steel. Well-defined passive regions are identified for all three conditions and the passivation range is approximately 270 mV until break down. The anodic dissolution behaviour of 800 °C and 1000 °C aged samples is fairly similar, which results in a slight difference of corrosion potential (E_{corr}) and the corrosion current density (I_{corr}). The sample aged at 1200 °C possesses both higher E_{corr} and smaller I_{corr} , which indicates that it may have better corrosion resistance than other two conditions. AISI 304L stainless steel exhibits a higher E_{corr} and a much larger passivation region (508 mV, from –189 to 319 mV) [49] compared to all AlCoCrFeNi specimens, making the corrosion resistance superior to AlCoCrFeNi.

Fig. 12 shows the surface corrosion of fracture that formed following electrochemical polarization of all the samples. A large extent of pitting corrosion is observed for each specimen. As the initiation of a pit is associated with the breakdown of the protective film on the metal surface, the observation is consistent with the presentation of passive region on the polarization curves and is evidence of the formation of passive film on the surface. 800 °C and 1000 °C aged samples exhibit a few small pits and some large pits, while 1200 °C aged sample shows mostly large pits. The surface corrosion on all specimens indicate that Cl^- ions severely attack the Al-Ni enriched B2 phase, while Fe-Cr enriched FCC phase remained largely un-attacked.

The corrosion preference of the B2 phase could be primarily

Table 3

Electrochemical parameters of aged AlCoCrFeNi specimens when polarized in 0.6 M NaCl solution at room temperature.

Ageing cycle	E_{corr} (mV)	I_{corr} ($\mu\text{A}/\text{cm}^2$)	E_{pit} (mV)
800 °C/168 h	–264.1	0.117	5.1
1000 °C/168 h	–259.6	0.129	–7.2
1200 °C/168 h	–240.9	0.089	35.5
SS304	–189	0.29	319

attributed to galvanic coupling of the FCC precipitate and the B2 matrix at micro scale, where the precipitated FCC acts as the cathode while the B2 matrix acts as the anode. When the sample was placed in the 0.6 M NaCl electrolyte solution, they are electrically connected and an electric potential exists. This potential difference provides a stronger driving force for the dissolution of the less noble Cr-depleted B2 matrix. As a result of the coupling effects between the matrix and second phase, the B2 matrix was eventually dissolved (corroded).

The corrosion properties of as cast bulk AlCoCrFeNi prepared by arc melting and AlCoCrFeNi coating prepared by electrospray deposition were studied [50]. It was found that unlike bulk sample, the coating showed no obvious chemical segregation and fine precipitate structure, leading to relatively uniform corrosion attack compared to the selective corrosion attack observed in the bulk sample. In the present study, samples aged 800 °C and 1000 °C exhibit larger volume fraction of FCC phase and much finer FCC precipitates than the sample aged at 1200 °C, which would contribute to a more substantial coupling effect resulting in an increased number of pits. That's might be the reason for a higher corrosion potential and smaller corrosion current density in 1200 °C aged condition.

4. Conclusion

1. The AlCoCrFeNi high entropy alloy as-deposited sample and the sample aged at 600 °C exhibit near single phase B2 solid solution structure, while the alloys aged at 800–1200 °C show FCC precipitates in a B2 matrix.

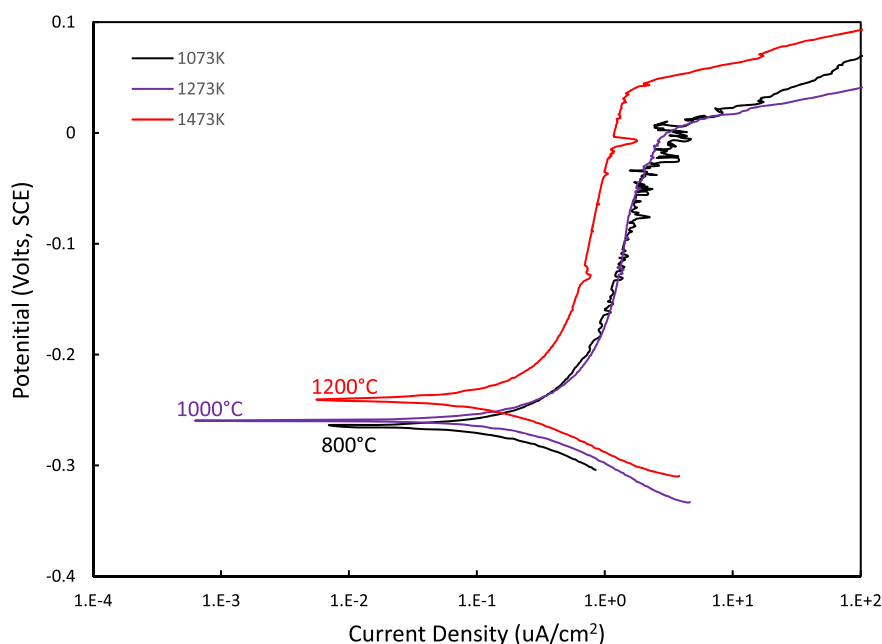


Fig. 11. Polarization diagrams for AlCoCrFeNi aged at 800 °C, 1000 °C and 1200 °C in 0.6 M NaCl solution at room temperature.

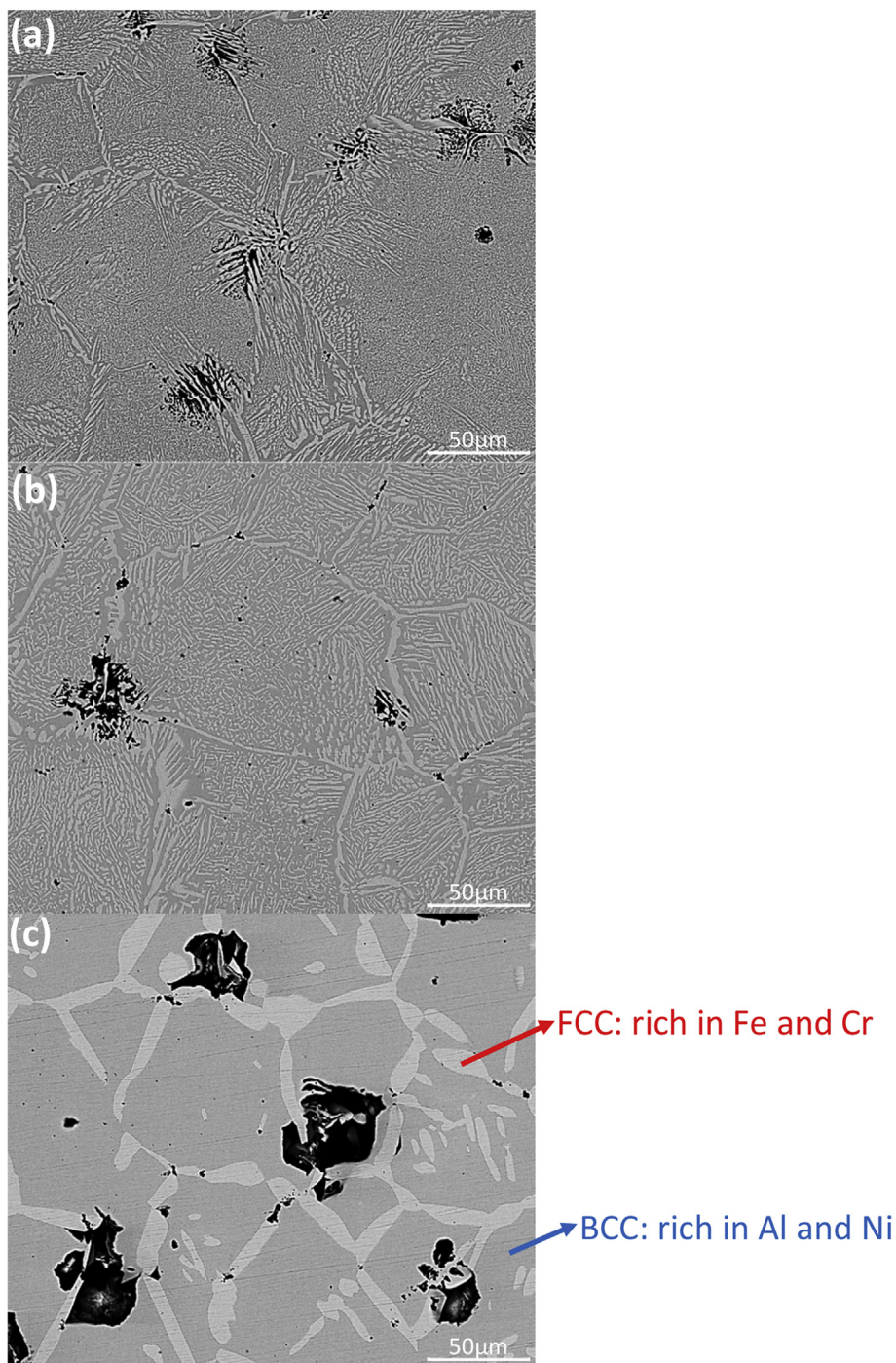


Fig. 12. SEM images of the corroded surfaces of AlCoCrFeNi aged at 800 °C (a), 1000 °C (b) and 1200 °C (c) after polarization test.

2. The as-deposited sample and sample aged at 600 °C exhibit periodic, fine scale structures consisting of alternating bright and dark interconnected B2 phases consistent with formation by spinodal decomposition.
3. Samples aged at 800 °C and 1000 °C exhibit needle shaped, plate shaped intergranular FCC precipitates and FCC precipitate at grain boundaries, while the sample aged at 1200 °C exhibits coarsened FCC precipitates mostly at grain boundaries.
4. Elemental segregation is observed for all specimens. The B2 phase is rich in Al and Ni while Fe and Cr segregate into FCC phase.
5. AlCoCrFeNi exhibits excellent compressive properties. After ageing, yield strength is reduced but ductility is improved, as softer FCC phase is formed. The higher ultimate fracture strength of the samples aged at higher temperatures corresponds to the increased ductility contributing to additional propensity for strain hardening. Based on the fractographs, the fracture mechanism is consistent with typical cleavage fracture except for 800 °C and 1000 °C aged specimen. The existence of FCC precipitates within the B2 grains in samples aged at 800 °C and 1000 °C results in a combination of cleavage and quasi-cleavage fracture mechanisms.

6. The corrosion potential and pitting potential of the AlCoCrFeNi alloy tested in 0.6 M NaCl is significantly lower than those of AISI 304L stainless steel. Pitting corrosion occurs following breakdown of the passive film. Al-Ni enriched B2 matrix is severely corroded as a result of galvanic coupling with Fe-Cr enriched FCC precipitate.

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References

- [1] P.K. Huang, J.W. Yeh, T.T. Shun, S.K. Chen, Multi-principal-element alloys with improved oxidation and wear resistance for thermal spray coating, *Adv. Eng. Mater.* 6 (2004) 74–78.
- [2] A. Gali, E.P. George, Tensile properties of high- and medium-entropy alloys, *Intermetallics* 39 (2013) 74–78.
- [3] C.-Y. Hsu, J.-W. Yeh, S.-K. Chen, T.-T. Shun, Wear resistance and high-temperature compression strength of fcc CuCoNiCrAl0.5Fe alloy with boron addition, *Metall. Mat. Trans. A* 35 (2004) 1465–1469.
- [4] M.A. Hemphill, T. Yuan, G.Y. Wang, J.W. Yeh, C.W. Tsai, A. Chuang, P.K. Liaw, Fatigue behavior of Al0.5CoCrCuFeNi high entropy alloys, *Acta Mater.* 60 (2012) 5723–5734.
- [5] Y.-F. Kao, T.-D. Lee, S.-K. Chen, Y.-S. Chang, Electrochemical passive properties of AlxCoCrFeNi ($x = 0, 0.25, 0.50, 1.00$) alloys in sulfuric acids, *Corros. Sci.* 52 (2010) 1026–1034.
- [6] Y. Zhang, T.T. Zuo, Z. Tang, M.C. Gao, K.A. Dahmen, P.K. Liaw, Z.P. Lu, Microstructures and properties of high-entropy alloys, *Prog. Mater. Sci.* 61 (2014) 1–93.
- [7] M.-H. Tsai, J.-W. Yeh, High-entropy alloys: a critical review, *Mater. Res. Lett.* 2 (2014) 107–123.
- [8] X. Yang, Y. Zhang, Prediction of high-entropy stabilized solid-solution in multi-component alloys, *Mater. Chem. Phys.* 132 (2012) 233–238.
- [9] S. Guo, Q. Hu, C. Ng, C.T. Liu, More than entropy in high-entropy alloys: forming solid solutions or amorphous phase, *Intermetallics* 41 (2013) 96–103.
- [10] S. Guo, C. Ng, J. Lu, C.T. Liu, Effect of valence electron concentration on stability of fcc or bcc phase in high entropy alloys, *J. Appl. Phys.* 109 (2011) 103505.
- [11] D.J.M. King, S.C. Middleburgh, A.G. McGregor, M.B. Cortie, Predicting the formation and stability of single phase high-entropy alloys, *Acta Mater.* 104 (2016) 172–179.
- [12] B. Cantor, High-entropy alloys, in: K.H.J. Buschow, R.W. Cahn, M.C. Flemings, B. Ilshner, E.J. Kramer, S. Mahajan, P. Veyssière (Eds.), *Encyclopedia of Materials: Science and Technology*, second ed., Elsevier, Oxford, 2011, pp. 1–3.
- [13] J.-W. Yeh, S.-K. Chen, J.-Y. Gan, L. Su-Jien, et al., Communications: formation of simple crystal structures in Cu-Co-Ni-Cr-Al-Fe-Ti-V alloys with multiprincipal metallic elements, *Metall. Mater. Trans.* 35A (2004) 2533–2536.
- [14] O.N. Senkov, J.M. Scott, S.V. Senkova, F. Meisenkothen, D.B. Miracle, C.F. Woodward, Microstructure and elevated temperature properties of a refractory TaNbHfZrTi alloy, *J. Mater. Sci.* 47 (2012) 4062–4074.
- [15] O.N. Senkov, G.B. Wilks, J.M. Scott, D.B. Miracle, Mechanical properties of Nb25Mo25Ta25W25 and V20Nb20Mo20Ta20W20 refractory high entropy alloys, *Intermetallics* 19 (2011) 698–706.
- [16] C.-Y. Hsu, C.-C. Juan, W.-R. Wang, T.-S. Sheu, J.-W. Yeh, S.-K. Chen, On the superior hot hardness and softening resistance of AlCoCrFeMo0.5Ni high-entropy alloys, *Mater. Sci. Eng. A* 528 (2011) 3581–3588.
- [17] W.-R. Wang, W.-L. Wang, S.-C. Wang, Y.-C. Tsai, C.-H. Lai, J.-W. Yeh, Effects of Al addition on the microstructure and mechanical property of AlxCoCrFeNi high-entropy alloys, *Intermetallics* 26 (2012) 44–51.
- [18] Y.-F. Kao, T.-J. Chen, S.-K. Chen, J.-W. Yeh, Microstructure and mechanical property of as-cast, -homogenized, and -deformed AlxCoCrFeNi ($0 \leq x \leq 2$) high-entropy alloys, *J. Alloys Compd.* 488 (2009) 57–64.
- [19] Z. Tang, O.N. Senkov, C.M. Parish, C. Zhang, F. Zhang, L.J. Santodonato, G. Wang, G. Zhao, F. Yang, P.K. Liaw, Tensile ductility of an AlCoCrFeNi multi-phase high-entropy alloy through hot isostatic pressing (HIP) and homogenization, *Mater. Sci. Eng. A* 647 (2015) 229–240.
- [20] S.G. Ma, Y. Zhang, Effect of Nb addition on the microstructure and properties of AlCoCrFeNi high-entropy alloy, *Mater. Sci. Eng. A* 532 (2012) 480–486.
- [21] S.G. Ma, J.W. Qiao, E.W. Huang, C.P. Chuang, P.K. Liaw, Y. Zhang, Microstructural characteristics and mechanical behaviors of AlCoCrFeNi high-entropy alloys at ambient and cryogenic temperatures, *Mater. Sci. Forum* 688 (2011) 419–425.
- [22] W.-R. Wang, W.-L. Wang, J.-W. Yeh, Phases, microstructure and mechanical properties of AlxCoCrFeNi high-entropy alloys at elevated temperatures, *J. Alloys Compd.* 589 (2014) 143–152.
- [23] M.-H. Chuang, M.-H. Tsai, W.-R. Wang, S.-J. Lin, J.-W. Yeh, Microstructure and wear behavior of AlxCo1.5CrFeNi1.5Ti high-entropy alloys, *Acta Mater.* 59 (2011) 6308–6317.
- [24] Z. Fu, W. Chen, S. Fang, D. Zhang, H. Xiao, D. Zhu, Alloying behavior and deformation twinning in a CoNiFeCrAl0.6Ti0.4 high entropy alloy processed by spark plasma sintering, *J. Alloys Compd.* 553 (2013) 316–323.
- [25] K.B. Zhang, Z.Y. Fu, J.Y. Zhang, W.M. Wang, S.W. Lee, K. Niihara, Characterization of nanocrystalline CoCrFeNiTiAl high-entropy solid solution processed by mechanical alloying, *J. Alloys Compd.* 495 (2010) 33–38.
- [26] S. Praveen, B.S. Murty, R.S. Kottada, Alloying behavior in multi-component AlCoCrCuFe and NiCoCrCuFe high entropy alloys, *Mater. Sci. Eng. A* 534 (2012) 83–89.
- [27] W. Ji, Z. Fu, W. Wang, H. Wang, J. Zhang, Y. Wang, F. Zhang, Mechanical alloying synthesis and spark plasma sintering consolidation of CoCrFeNiAl high-entropy alloy, *J. Alloys Compd.* 589 (2014) 61–66.
- [28] Y.-L. Chen, C.-W. Tsai, C.-C. Juan, M.-H. Chuang, J.-W. Yeh, T.-S. Chin, S.-K. Chen, Amorphization of equimolar alloys with HCP elements during mechanical alloying, *J. Alloys Compd.* 506 (2010) 210–215.
- [29] X.-W. Qiu, Microstructure and properties of AlCrFeNiCoCu high entropy alloy prepared by powder metallurgy, *J. Alloys Compd.* 555 (2013) 246–249.
- [30] L.M. Wang, C.C. Chen, J.W. Yeh, S.T. Ke, The microstructure and strengthening mechanism of thermal spray coating NiCo0.6Fe0.2CrSiAlTi0.2 high-entropy alloys, *Mater. Chem. Phys.* 126 (2011) 880–885.
- [31] S.-Y. Chang, S.-Y. Lin, Y.-C. Huang, C.-L. Wu, Mechanical properties, deformation behaviors and interface adhesion of (AlCrTaTiZr)Nx multi-component coatings, *Surf. Coat. Technol.* 204 (2010) 3307–3314.
- [32] X. Ye, M. Ma, W. Liu, L. Li, M. Zhong, Y. Liu, Q. Wu, Synthesis and characterization of high-entropy alloy AlXFeCoNiCuCr by laser cladding, *Adv. Mater. Sci. Eng.* 2011 (2011).
- [33] H. Zhang, Y. Pan, Y. He, Effects of annealing on the microstructure and properties of 6FeNiCoCrAlTiSi high-entropy alloy coating prepared by laser cladding, *J. Therm. Spray. Tech.* 20 (2011) 1049–1055.
- [34] D.A. Hollander, M. von Walter, T. Wirtz, R. Sellei, B. Schmidt-Rohlfing, O. Paar, H.-J. Erli, Structural, mechanical and in vitro characterization of individually structured Ti–6Al–4V produced by direct laser forming, *Biomaterials* 27 (2006) 955–963.
- [35] X. Wu, J. Liang, J. Mei, C. Mitchell, P.S. Goodwin, W. Voice, Microstructures of laser-deposited Ti–6Al–4V, *Mater. Des.* 25 (2004) 137–144.
- [36] F. Wang, J. Mei, X. Wu, Microstructure study of direct laser fabricated Ti alloys using powder and wire, *Appl. Surf. Sci.* 253 (2006) 1424–1430.
- [37] F. Wang, J. Mei, X. Wu, Direct laser fabrication of Ti6Al4V/TiB, *J. Mater. Process. Technol.* 195 (2008) 321–326.
- [38] K.I. Schwendner, R. Banerjee, P.C. Collins, C.A. Brice, H.L. Fraser, Direct laser deposition of alloys from elemental powder blends, *Scr. Mater.* 45 (2001) 1123–1129.
- [39] P.C. Collins, R. Banerjee, H.L. Fraser, The influence of the enthalpy of mixing during the laser deposition of complex titanium alloys using elemental blends, *Scr. Mater.* 48 (2003) 1445–1450.
- [40] F. Zhang, J. Chen, H. Tan, X. Lin, W. Huang, Composition control for laser solid forming from blended elemental powders, *Opt. Laser Technol.* 41 (2009) 601–607.
- [41] I. Kunce, M. Polanski, K. Karczewski, T. Plocinski, K.J. Kurzydowski, Microstructural characterisation of high-entropy alloy AlCoCrFeNi fabricated by laser engineered net shaping, *J. Alloys Compd.* 648 (2015) 751–758.
- [42] I. Kunce, M. Polanski, J. Bystrzycki, Microstructure and hydrogen storage properties of a TiZrNbMoV high entropy alloy synthesized using Laser Engineered Net Shaping (LENS), *Int. J. Hydrogen Energy* 39 (2014) 9904–9910.
- [43] J. Joseph, T. Jarvis, X. Wu, N. Stanford, P. Hodgson, D.M. Fabijanic, Comparative study of the microstructures and mechanical properties of direct laser fabricated and arc-melted AlxCoCrFeNi high entropy alloys, *Mater. Sci. Eng. A* 633 (2015) 184–193.
- [44] Y.J. Zhou, Y. Zhang, Y.L. Wang, G.L. Chen, Solid solution alloys of AlCoCrFeNiTx with excellent room-temperature mechanical properties, *Appl. Phys. Lett.* 90 (2007).
- [45] C. Zhang, F. Zhang, S. Chen, W. Cao, Computational thermodynamics aided high-entropy alloy design, *JOM* 64 (2012) 839–845.
- [46] Z. Tang, M. Gao, H. Diao, T. Yang, J. Liu, T. Zuo, Y. Zhang, Z. Lu, Y. Cheng, Y. Zhang, K. Dahmen, P. Liaw, T. Egami, Aluminum alloying effects on lattice types, microstructures, and mechanical behavior of high-entropy alloys systems, *JOM* 65 (2013) 1848–1858.
- [47] S.C. Middleburgh, D.M. King, G.R. Lumpkin, M. Cortie, L. Edwards, Segregation and migration of species in the CrCoFeNi high entropy alloy, *J. Alloys Compd.* 599 (2014) 179–182.
- [48] C.R. Brooks, *Failure Analysis of Engineering Materials*, McGraw-Hill, New York, 2002.
- [49] T. Balusamy, T.S.N. Sankara Narayanan, K. Ravichandran, I.S. Park, M.H. Lee, Influence of surface mechanical attrition treatment (SMAT) on the corrosion behaviour of AISI 304 stainless steel, *Corros. Sci.* 74 (2013) 332–344.
- [50] Q.H. Li, T.M. Yue, Z.N. Guo, X. Lin, Microstructure and corrosion properties of AlCoCrFeNi high entropy alloy coatings deposited on AISI 1045 steel by the electrospray process, *Metall. Mat. Trans. A* 44 (2013) 1767–1778.